

DermaNet: Flexible-View Fusion for Dermoscopic–Clinical Skin Lesion Classification

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Challenges

- **Fine-grained classification:** Small images with limited spatial structure
- **Imbalanced classes:** Rare classes amount to $\leq 1\%$ of the dataset
- **Out-of-Distribution (OOD) detection:** Important to detect samples from unknown classes
- **Uncertainty estimation:** Important for medical applications and OOD detection
- **Two views/modalities:** Paired dermoscopic and clinical images in MILK10
- **Explainability:** Identifying clinical features, displaying attention maps, etc.



Dermoscopic Image



Clinical Image

ISIC2029

- **Dermoscopic images only**
- 8 classes + OTHER in test set
- Train set: 25,331 images
- Held-out test set: 8,238 images (+OTHER class)

MILK10

- **Paired clinical + dermoscopic images**
- 9+2 OTHER classes
- Train set: 10,480 paired images
- Held-out test set: 958 paired images

Class	Training %
NV (Melanocytic nevus)	50.8%
MEL (Melanoma)	17.9%
BCC (Basal cell carcinoma)	13.1%
BKL (Benign keratosis)	10.4%
AK (Actinic keratosis)	3.4%
SCC (Squamous cell carcinoma)	2.5%
VASC (Vascular lesion)	1.0%
DF (Dermatofibroma)	0.9%

Class	Training %
BCC (Basal cell carcinoma)	48.1%
NV (Melanocytic nevus)	14.2%
BKL (Benign keratinocytic les.)	10.4%
SCC/KA (Sq.cell carc./KA)	9.0%
MEL (Melanoma)	8.6%
AK/IEC (Actinic keratosis/IEC)	5.8%
DF (Dermatofibroma)	1.0%
INF (Inflammatory/infectious)	1.0%
VASC (Vascular lesion)	0.9%
BEN_OTH (Other benign)	0.8%
MAL_OTH (Other malignant)	0.2%

Baseline Architectures for ISIC2019

- Comparison of **EfficientNet-B5**, **ViT**, **MambaVision** on the official test set
- Focal loss, test-time augmentation, and hyperparameter optimization
- Heterogeneous ensemble of 3 complementary architectures:

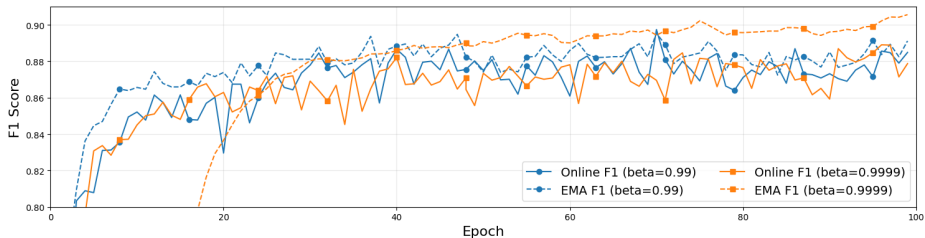
→ EfficientNet-B5: 73.6%, MambaVision: 72.3%, ViT: 72.2%

→ **74.3% Weighted-F1** → **74.8% Accuracy**

Aktan et al., "Comparison of Leading Deep Learning Architectures on Skin Lesion Classification Using the ISIC 2019 Dataset" TIPTEKNO 2025, Gazi Mağusa, Northern Cyprus.

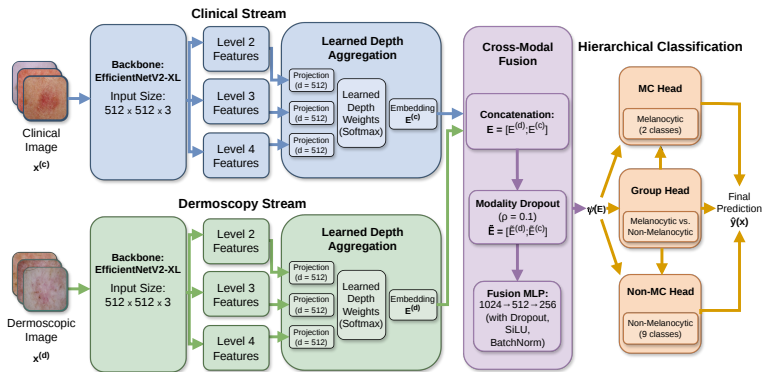
Reducing parameter fluctuations during convergence

- Parameter instability due to small-batch training
- Student: Swin V2 trained on **ISIC2019**
- Teacher keeps track of the Exponential Moving Average (**EMA**) of online model
- No additional inference cost!



→ +4.17% Macro-F1 → 60.18% to 64.35%

DermaNet: Overall Architecture



Multi-Level
Features

Paired Feature
Fusion

Hierarchical
Prediction

Flexible View
(Paired or Single)

1. Multi-Level Representation Learning

Multilevel Features

We extract features from three depths:

$$L_2, \quad L_3, \quad L_4$$

Each layer's representation is separately projected to a 512-dimensional representation e_i , $i \in 2, 3, 4$.

Learned depth aggregation

The weight of each layer is **learned**:

$$\alpha = \text{softmax}(w)$$

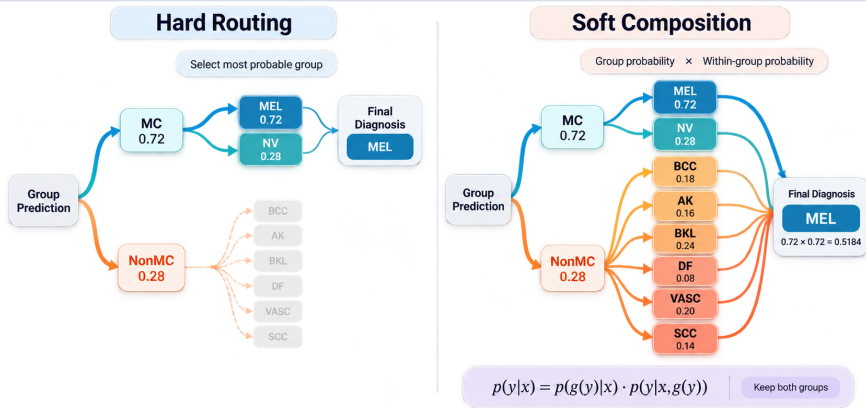
Final representation:

$$E = \alpha_2 e_2 + \alpha_3 e_3 + \alpha_4 e_4$$

→ **+0.90 Macro-F1 in MILK10** → **+1.55 Micro-F1 in ISIC2019**

compared to baseline models (EfficientNet-B5 using just the last embedding layer).

2. Clinically Structured Hierarchical Classification



→ +4.2 Macro-F1 in MILK10 → + 2.2 Micro-F1 in ISIC2019
compared to EfficientNet-B5 with multi-level features.

3. Paired (Clinical-Dermoscopic) Feature Fusion

Separate view representations

After multi-level aggregation:

$$E^{(c)} \in \mathbb{R}^{512}$$

for the clinical image, and

$$E^{(d)} \in \mathbb{R}^{512}$$

for the dermoscopic image.

Joint representation

The two embeddings are concatenated:

$$E = [E^{(c)}; E^{(d)}] \in \mathbb{R}^{1024}$$

and processed by the fusion network:

$$1024 \rightarrow 512 \rightarrow 256$$

The final 256-dimensional representation is shared by all classification heads.

→ **+4.3 Macro-F1 in MILK10** compared to EfficientNet-B5 with multi-level features and hierarchical prediction.

4. Flexible-View Learning with Modality Dropout

Training Condition	Clinical	Dermoscopy	Probability
Both views	✓	✓	80%
Clinical only	✓	×	10%
Dermoscopy only	×	✓	10%

→ Modality drop out enables training and inference with paired or single views.

Ablation Study

Configuration	MILK10k		ISIC2019	
	Macro-F1	Δ	Micro-F1	Δ
Baseline EfficientNetV2	40.3%	–	73.63%	–
+ Multi-Level Aggregation	41.2%	+0.9	75.18%	+1.55
+ Hierarchical Prediction	45.4%	+4.2	77.42%	+2.24
+ Paired Feature Fusion	49.7%	+4.3	–	–

MILK10k

Largest gains came from **paired feature fusion** and **hierarchical prediction**.

40.3% → **49.7%**

ISIC2019

textbfHierarchical prediction and **multi-level aggregation** improved performance.

73.63% → **77.42%**

ISIC2019: Comparison with Prior Work

70:10:20 Train–Validation–Test Split

Method	Micro-F1	Macro-F1	AUC	Bal. Acc.
Sun et al. (2021)	–	–	91.5%	66.2%
Yao et al. (2022)	75.0%	–	–	–
Li et al. (2022)	75.1%	–	–	–
Alsahafi et al. (2023)	71.3%	–	–	–
Liu et al. (2026)	79.9%	68.9%	–	–
DermaNet	82.6%	72.7%	95.4%	70.8%

DermaNet achieves strong performance across all reported metrics under the same 70:10:20 evaluation setting.